

2025-P3-Science-End Of Year Exam-Nanyang

1 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #1)

James made the following observations about his dog.

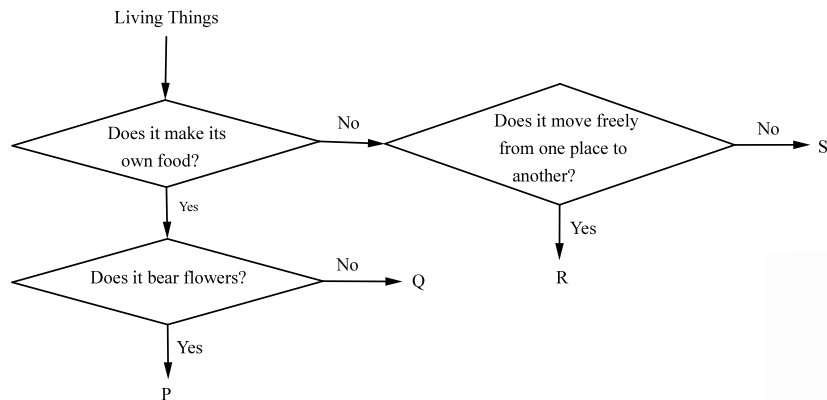
- A** It gave birth to puppies.
- B** It shivers when feeling cold.
- C** It became too heavy to be carried.

Which of the observations show that James' dog is a living thing?

- A. **A** and **B** only B. **A** and **C** only C. **B** and **C** only D. **A**, **B** and **C**

2 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #2)

Study the diagram below.



Which of the following shows the living things represented by **P**, **Q**, **R** and **S**?

	P	Q	R	S
(1)	moss	mushroom	hibiscus plant	cat
(2)	mushroom	moss	hibiscus plant	cat
(3)	hibiscus plant	moss	cat	mushroom
(4)	hibiscus plant	mushroom	cat	moss

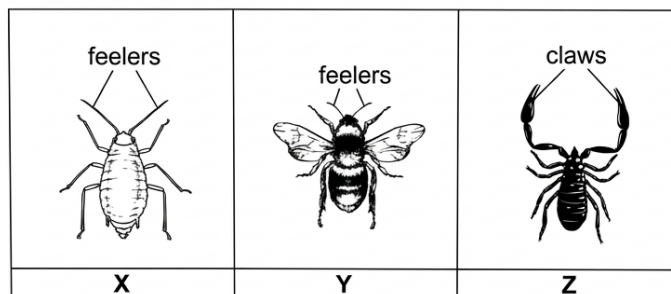
3 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #3)

Which statement about fish and reptiles is correct?

- A. Both have fins.
- B. Both are covered with scales.
- C. Fish lay eggs but reptiles give birth to young.
- D. Fish live in water but reptiles live only on land.

4 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #4)

Three animals, *X*, *Y* and *Z* are shown below.

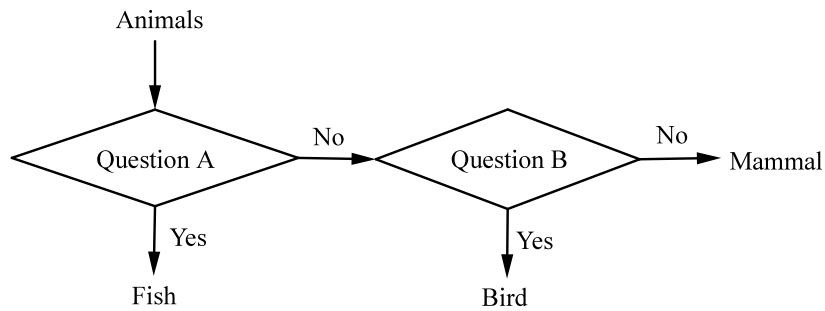


Based on the diagrams above, which animals can be correctly classified as insects?

- A. *X* and *Y* only
- B. *X* and *Z* only
- C. *Y* and *Z* only
- D. *X*, *Y* and *Z*

5 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #5)

Study the chart below.



Which of the following represents Questions *A* and *B* correctly?

	Question A	Question B
(1)	Does it have gills?	Does it have hair?
(2)	Does it have a tail?	Does it have wings?
(3)	Does it have scales?	Does it have four legs?
(4)	Does it have fins?	Does it have feathers?

6 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #6)

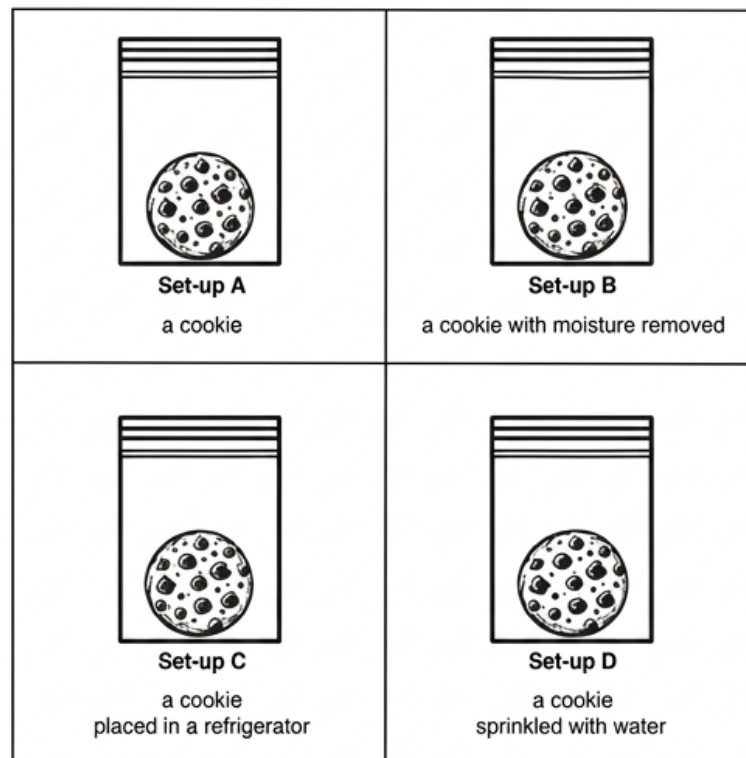
Which of the following statements are true about bacteria?

- A* Bacteria are living things.
- B* All bacteria are harmful to humans.
- C* Bacteria come in different shapes and sizes.
- D* We can easily see bacteria with our naked eye.

A. *A* and *B* only B. *A* and *C* only C. *B* and *D* only D. *C* and *D* only

7 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #7)

Lindy placed 4 cookies in 4 identical sealed bags in a room as shown below.



In which set-up would mould most likely grow on the cookie first?

- A. Set-up *A* B. Set-up *B* C. Set-up *C* D. Set-up *D*

8 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #8)

Which of the following is correct for both the life cycle of a frog and the life cycle of a cockroach?

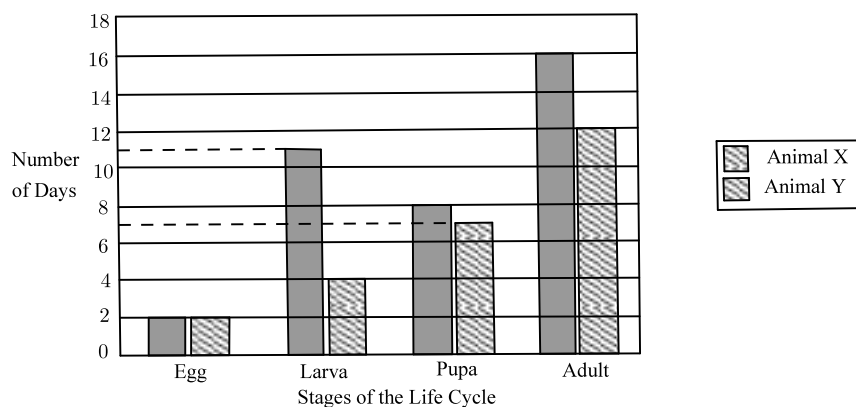
- A Both the young look like the adult.
 B Both life cycles take place fully on land.
 C Both life cycles have the same number of stages.

- A. *A* only B. *C* only C. *A* and *B* only D. *B* and *C* only

9

(SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #9)

The graph below shows how long Animal X and Animal Y spent in each stage of their life cycle.

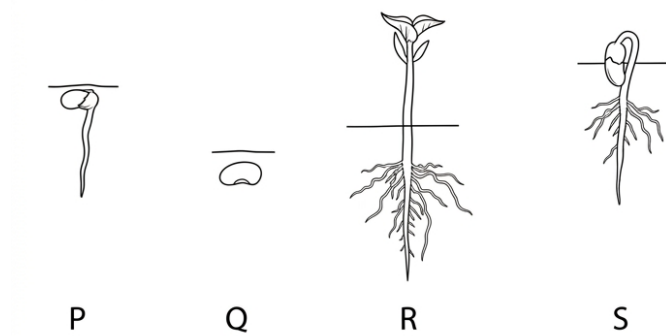


At which stage will Animal *X* and Animal *Y* be on the 12th day after the eggs hatch?

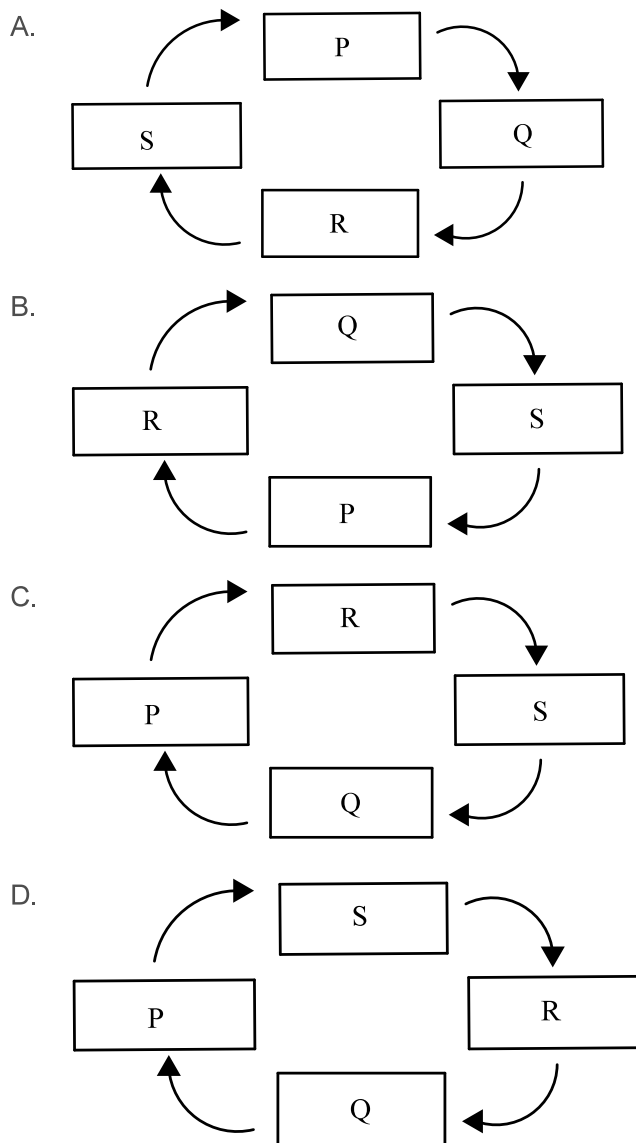
	Animal X	Animal Y
(1)	larva	adult
(2)	pupa	larva
(3)	pupa	adult
(4)	adult	pupa

10 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #10)

The diagram below shows the various stages in the life cycle of a plant.

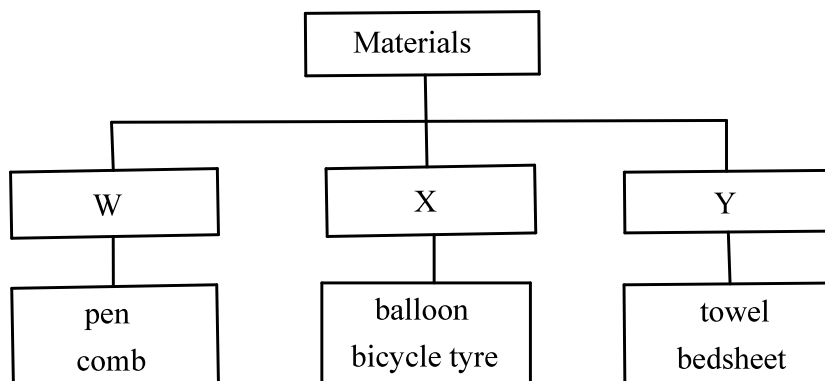


Which one of the following correctly shows the life cycle of a plant?



11 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #11)

Study the diagram below.



What could materials *W*, *X* and *Y* be?

	W	X	Y
(1)	metal	wood	fabric
(2)	wood	glass	plastic
(3)	plastic	rubber	fabric
(4)	ceramic	rubber	metal

12 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #12)

Study the diagram below.

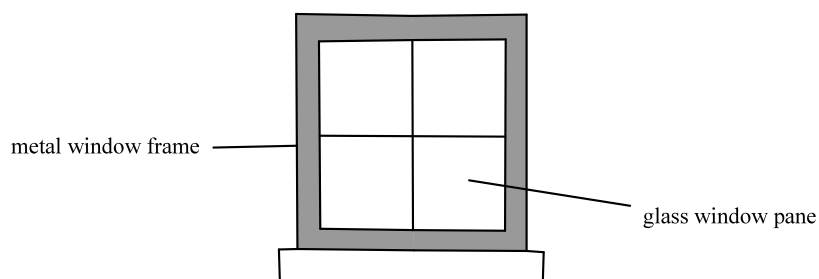
Group <i>A</i>	Group <i>B</i>
metal block glass marble	balloon filled with air styrofoam ball

How have the objects been classified into Group *A* and *B*?

	Group A	Group B
(1)	flexible	not flexible
(2)	sink in water	float in water
(3)	break easily	do not break easily
(4)	allow most light to pass through	do not allow light to pass through

13 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #13)

Study the diagram below. Different materials are used to make different parts of a window.

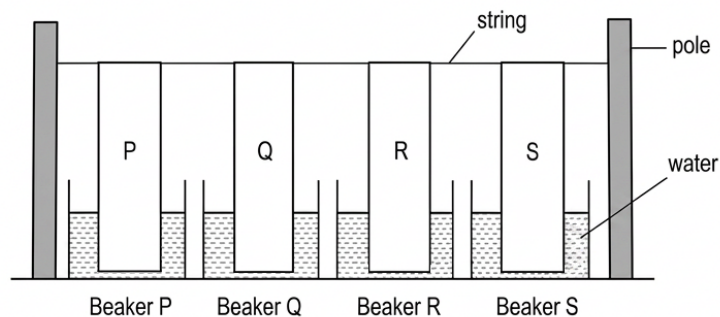


Which of the following best describes the properties of materials used to make the different parts of the window?

	metal	glass
(1)	strong	allows light to pass through
(2)	flexible	does not allow light to pass through
(3)	waterproof	flexible
(4)	able to float on water	waterproof

14 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #14)

Ben wanted to find out which materials, *P*, *Q*, *R*, or *S*, would make the best raincoat. He hung a similar piece of each material on a string. The materials were dipped in **20ml** of water.



After ten minutes, Ben removed each material from the beaker and measured the amount of water left. He recorded the results in the table below.

	Material			
	P	Q	R	S
Amount of water at the start (ml)	20	20	20	20
Amount of water left in each beaker (ml)	6	0	20	13

Which is the best material to make the raincoat?

A. *P*

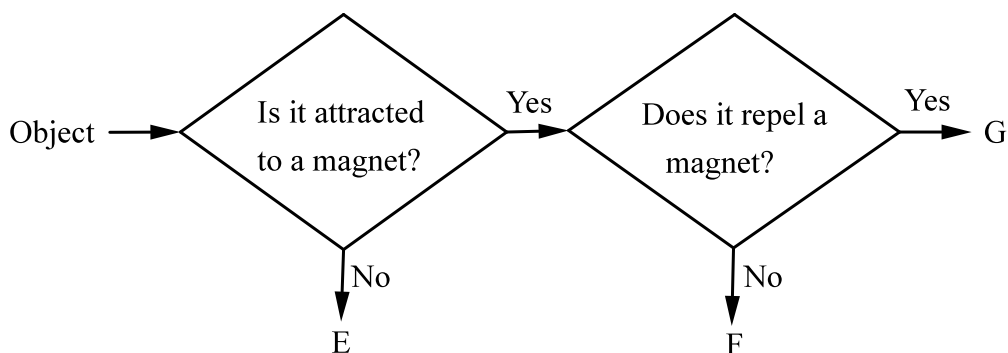
B. *Q*

C. *R*

D. *S*

15 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #15)

Study the flow chart below carefully.

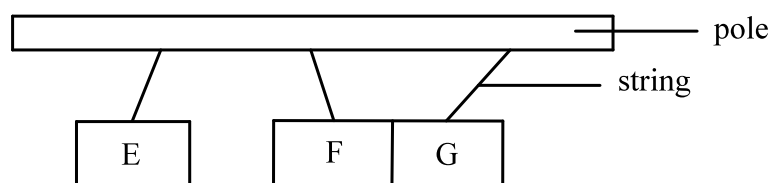


What could object *E*, *F* and *G* be?

	Object E	Object F	Object G
(1)	wooden ruler	bar magnet	iron nail
(2)	steel keychain	wooden ruler	button magnet
(3)	plastic spoon	iron rod	ring magnet
(4)	iron nail	plastic spoon	steel keychain

16 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #16)

Lucas hung three bars, *E*, *F* and *G* on a pole. The diagram below shows the interaction of the bars when they are brought near one another.



Which of the statements are true?

A Bar *F* is a magnet

B Bar *E* is made of plastic

C Bar *G* could be a magnetic material

A. *A* and *B* only

B. *A* and *C* only

C. *B* and *C* only

D. *A*, *B* and *C*

17 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #17)

Annie brought Object *X* near Object *Y*.

She observed that Object *Y* got pushed away.



Which one of the following statements could be true?

- A. Only Object *X* is a magnet.
- B. Only Object *Y* is a magnet.
- C. Both Object *X* and *Y* are magnets.
- D. Both Object *X* and *Y* are not magnets.

18 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #18)

Serena is lost in a forest. She found a small bar magnet in her bag and decides to suspend it freely using a thread.

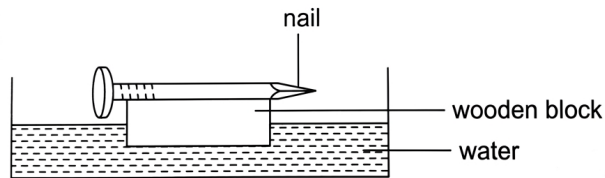
After a few minutes, the magnet stopped spinning and pointed in a specific direction.

Which one of the following best explains the direction that the magnet is pointing in?

- A. It is pointing towards a metal.
- B. It is pointing towards the sunlight.
- C. It is pointing in the East - South direction.
- D. It is pointing in the North - South direction.

19 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #19)

Rahim stroked an iron nail many times with a strong magnet. He then placed the nail on top of a wooden block. He placed the nail and wooden block in a bowl of water and noticed that they float in a certain direction.



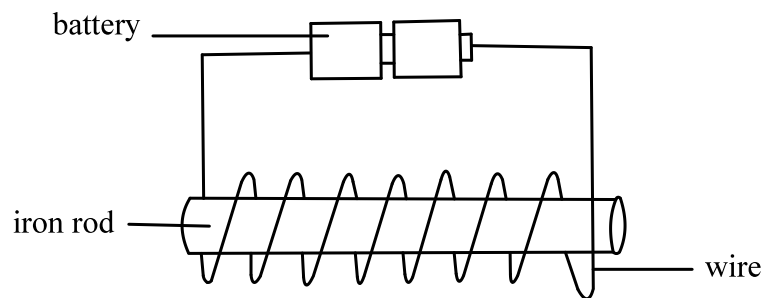
What was Rahim most likely trying to do?

- A* He wanted to find the direction of the wind.
- B* He wanted to find which direction was North.
- C* He wanted to find out if wood floats on water.
- D* He wanted to find out the strength of the iron nail.

A. *B* only B. *A* and *D* only C. *B* and *C* only D. *A*, *C* and *D* only

20 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #20)

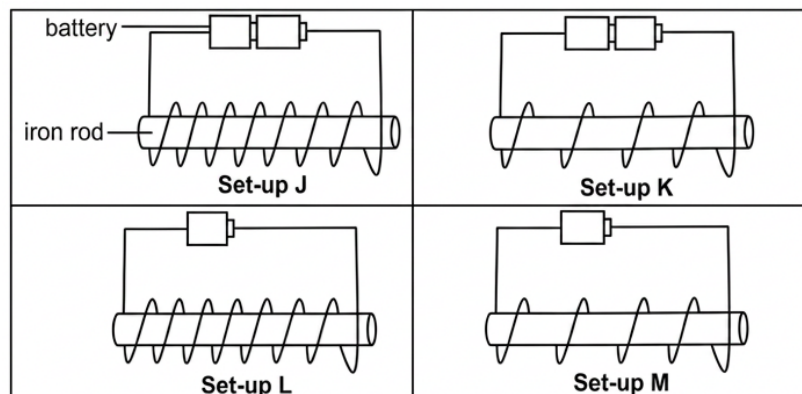
Which one of the following methods can be used to increase the magnetic strength of an electromagnet as shown in the diagram below?



- A. Heating up the iron rod.
- B. Hammering the iron rod.
- C. Increasing the number of batteries used.
- D. Decreasing the number of coils of wire around the iron rod.

21 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #21)

Viki conducted an experiment using four set-ups J, K, L and M as shown below.



Which two set-ups should Viki use if she wants to find out whether the number of coils of wire affects the magnetic strength of the electromagnet?

- A. *J* and *L* B. *K* and *L* C. *K* and *M* D. *L* and *M*

22 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 1, Question #22)

Which of the following is a common use of magnets in everyday life?

- A* Attracting plastic objects
B Pouring water into a metal cup
C A refrigerator door that stays shut

- A. *A* only B. *C* only C. *A* and *B* only D. *B* and *C* only

23 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 2, Question #23)

Study the classification table below. The diagrams are not drawn to scale.

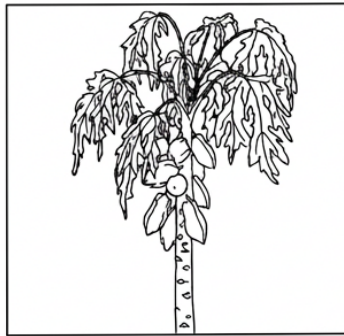
(1) Give suitable headings for each group by filling in the boxes below.

Things	
(i) _____	(ii) _____
 mould	 raincoat
 fern	 computer
 rabbit	 pillow

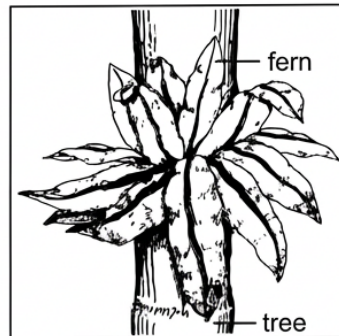
(2) Besides your answer in (1), state a difference in characteristic between the mould and the raincoat.

24 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 2, Question #24)

Meiling observed a papaya tree and a fern that grew on a tree.



Papaya tree



Fern that grew on a tree

- (1) State the way that they reproduce.
 - (i) papaya tree:
 - (ii) fern:
- (2) Meiling says that the fern gets its food from the tree that it grows on.
Give a reason why she is wrong.

25 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 2, Question #25)

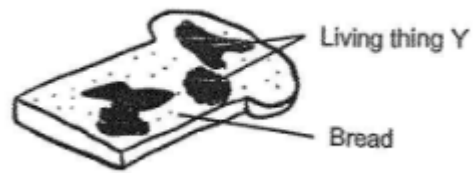
The diagram below shows Animal *S* which lives on land and in water. It has four legs and a moist skin.



- (1) Which animal group does Animal *S* belong to?
- (2) How does the moist skin help Animal *S* to stay alive in water?

26 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 2, Question #26)

Study the diagram below.

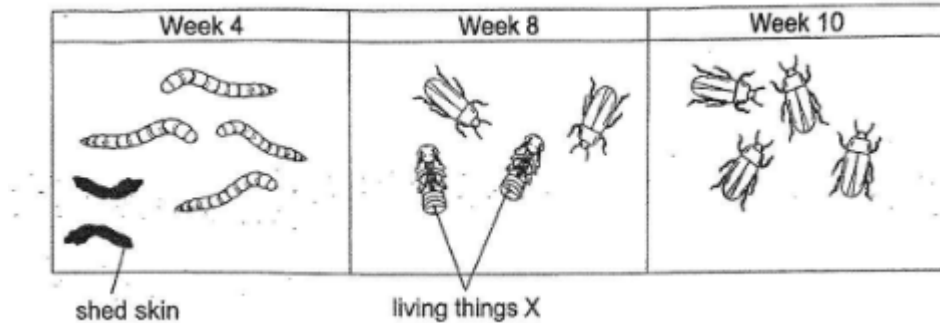


- (1) How does living thing *Y* reproduce?
- (2) How does living thing *Y* obtain food?

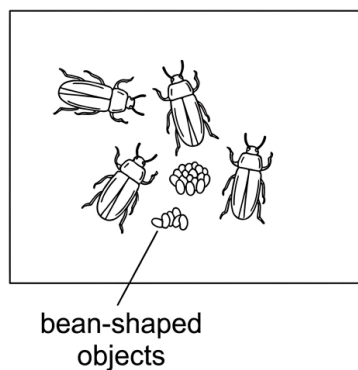
27 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 2, Question #27)

Ben kept the young of four insects with food in a container for ten weeks.

He drew his observations as shown in the table below.



- (1) At week 4, he noticed that there were some shed skin inside the container.
Why did the insects need to shed their skin?
- (2) At week 8, he could not find the same insects he had at week 4. In the container, there were two beetles and two living things *X* that did not move from place to place.
What could living thing *X* be?
- (3) Besides not moving from place to place, state another difference in characteristic between living thing *X* and the young in week 4.
- (4) After some time, some bean-shaped objects appeared in the container even though Ben did not add them.



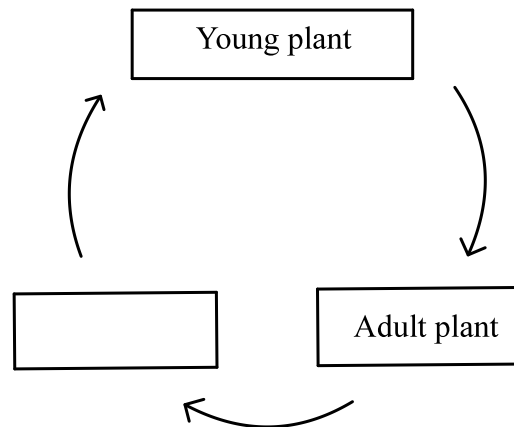
What had most likely happened for the bean-shaped objects to appear?

- (5) Name another animal that has a life cycle with the same number of stages as the insect that Ben kept.

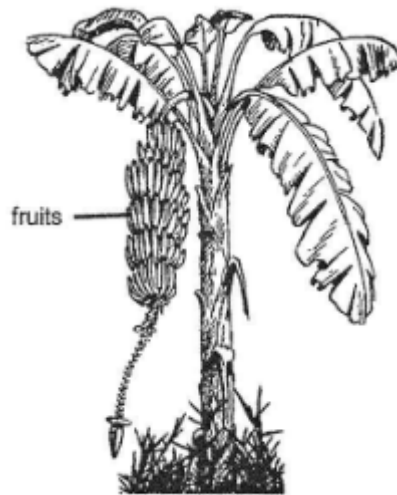
28 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 2, Question #28)

The diagram below shows the life cycle of a plant.

- (1) Fill in the correct stage in the box below.



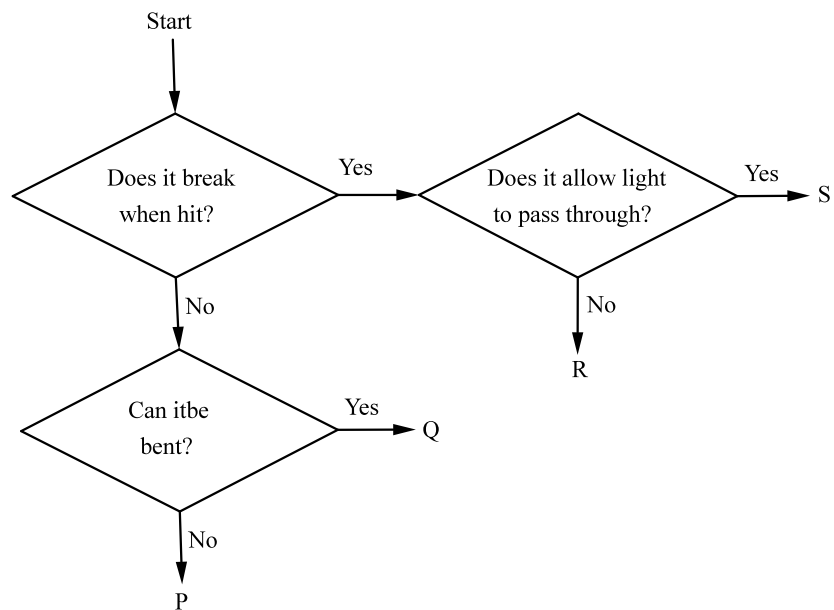
- (2) The diagram below shows a banana plant.



Which stage of the life cycle does the banana plant represent? Give a reason for your answer.

29 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 2, Question #29)

Study the flowchart.



- (1) State a difference in properties between objects *Q* and *R*.
- (2) Ben was given two objects, a metal hammer and glass spectacle lens.

Which letter, *P*, *Q*, *R* or *S*, best represent the two objects?

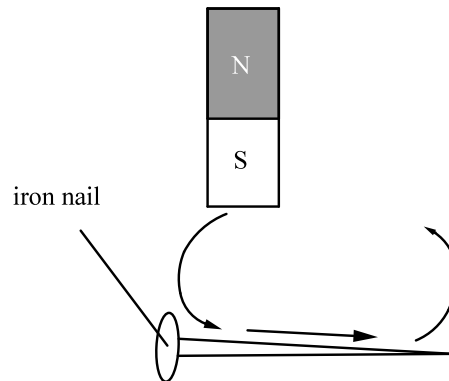
Spectacle lens: Letter _____

Metal hammer: Letter _____

30 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 2, Question #30)

Jimmy made a magnet using the stroke method as shown below.

He stroked the iron nail in the same direction with the same pole of a magnet.



- (1) The table below shows the number of steel clips attracted when different number of strokes are made.

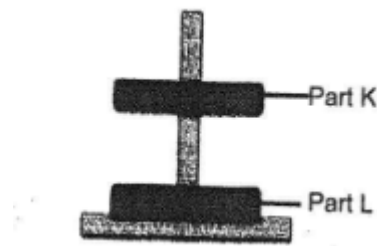
Number of Strokes made	Number of Steel Clips Attracted
10	2
20	9
30	17

Based on the results above, what is the relationship between the number of strokes made and the strength of the magnet?

- (2) Without changing the type of nail or the paper clips, suggest what Jimmy should do if he wanted the iron nail to attract more than 17 steel clips.
- (3) If Jimmy replaced the iron nail with a wooden nail and stroked it 30 times in the same direction with the same pole of a magnet, how many steel clips would it attract? Explain your answer.

31 (SG 2025-P3-Science-End Of Year Exam-Nanyang, paper 2, Question #31)

Jeslyn played with the toy shown below.

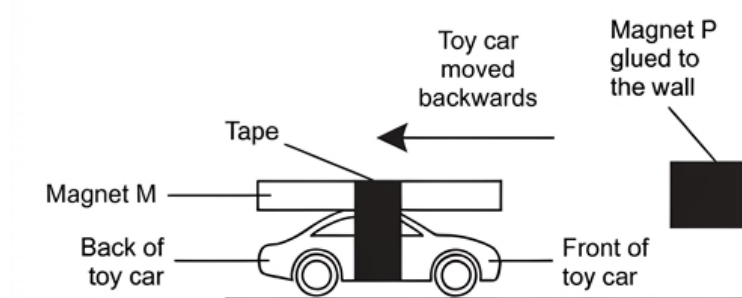


- (1) She observed that part *K* is 'floating' above part *L*.

Identify the objects that parts *K* and *L* are made of .

- (2) Jeslyn attached bar magnet *M* to the top of a toy car using some tape and rolled the toy car towards another magnet, *P*, that was glued onto the wall.

Before the front of the toy car reached the magnet *P*, Jeslyn observed that it started to move backwards.



Explain why the toy car moved backwards.